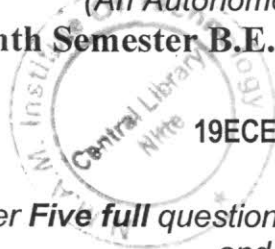


NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Eighth Semester B.E. (E&C) (Credit System) Degree Examinations
 May - June 2023



19ECE13 - MULTIMEDIA COMMUNICATIONS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Explain the role of data networks in multimedia communication.	10	L*2	1	1
	b) Discuss the interactive applications over the internet.	10	L2	2	1
2.	a) Explain the communication modes available to transfer the information stream.	10	L2	1	1
	b) With an example explain an ATM broadband multiservice network.	10	L2	2	1
3.	a) Explain the QoS parameters associated with: i) Packet-switched network. ii) Circuit-switched network.	12	L2	2	1
	b) Determine the propagation delay associated with the following communication channels: i) A connection through a private telephone network of 3km. ii) A connection through a PSTN of 300km. iii) A connection over a satellite channel of 40,000km. Assume that the velocity of propagation of a signal in the case of (i) and (ii) is $3 \times 10^8 \text{ ms}^{-1}$ and in the case of (iii) is $4 \times 10^8 \text{ ms}^{-1}$.	8	L3	2	1
Unit – II					
4.	a) With the aid of block diagram, explain LPC signal encoding and decoding principle.	10	L2	3	1
	b) Explain the implementation schematic and FIFO buffer operation for H.261 video encoder.	10	L2	3	1
5.	a) Explain the working principles of encoder and decoder for MPEG-4 standard.	10	L2	3	1
	b) Discuss the composition and format for MPEG-1 video bitstream structure.	10	L2	3	1
6.	a) Explain the usage of Ring networks to carry multimedia traffic.	10	L2	4	1
	b) Explain the working of a QoS processing model.	10	L2	4	1
Unit – III					
7.	a) Explain the system architecture and basic call flow associated with session initiation protocol.	10	L2	5	1
	b) Explain the usage of Real-time transport control protocol.	10	L2	5	1
8.	a) Write short notes on Session description protocol.	10	L2	5	1
	b) Write short notes on Real-time transport protocol.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
